

Self-Introduction

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- **Education:**

Undergraduate Student in Biomedical Engineering, Taipei Medical University

Double Major in Electrical Engineering , National Taipei University of Technology

- **Research Interest:**

Circuits designing , embedded systems , intelligent biomedical devices

- **Skills:**

Electronics: Circuit Analysis, Power Electronics (IsSpice), Control Systems

Signal & Data: Biomedical Signal Processing (MATLAB), Digital Signal Processing, Medical Imaging (MRI)

Programming: C / C++, x86 Microprocessor, Computer Networks, Logic Design

- **Experience:**

TMU Research Competition, 2025

NCKU MedTech Bootcamp, 2025

Smart TMU Workshop, 2024

Bio-ICT Project, 2023



Internship experience in AeProMED Innovation Co., Ltd. , 2025 summer



Electrocardiogram and cardiac output impedance signals in circuits

Presenter: Chang, Po-Hsiang

Advisor: Prof. Liu, Chien-chun

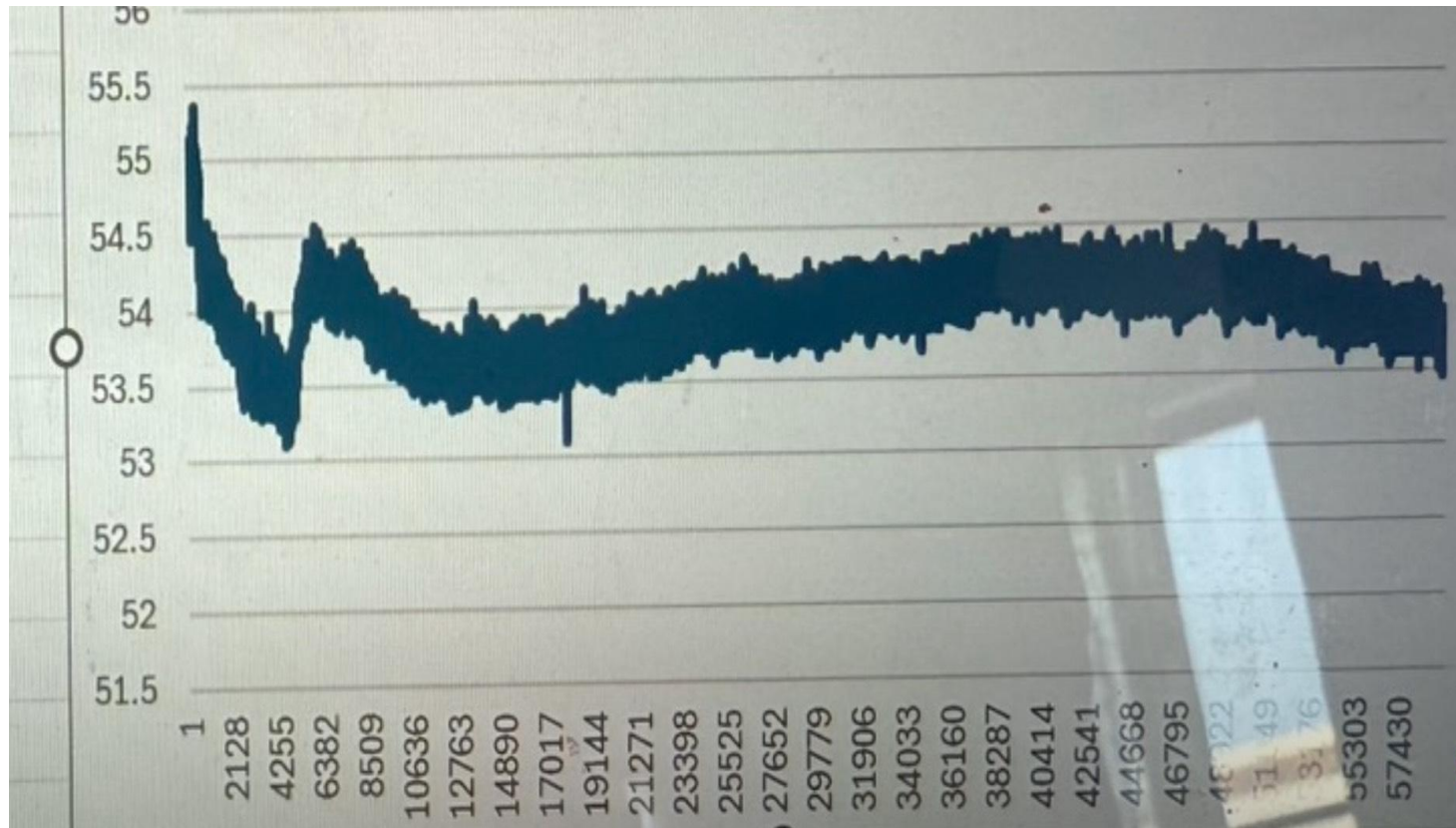


臺北醫學大學
TAIPEI MEDICAL UNIVERSITY

Contents

- **Bioelectrical Impedance**
 - Impedance cardiogram (ICG)
 - Equivalent circuit model of cell
 - AFE for VSM device
- **Electrocardiogram (ECG)**
 - ECG signal introduction
 - Circuit Design
 - Improvement Directions

ICG Introduction



Bioelectrical Impedance Equivalent CKT

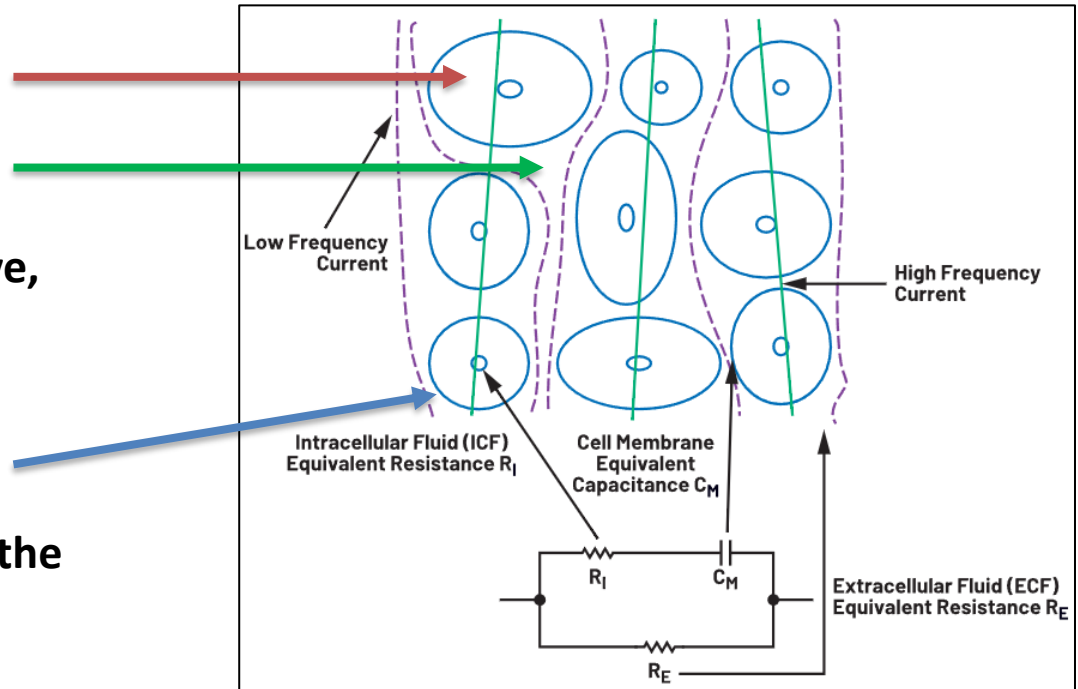
Intracellular fluid (ICF)

Extracellular fluid (ECF)

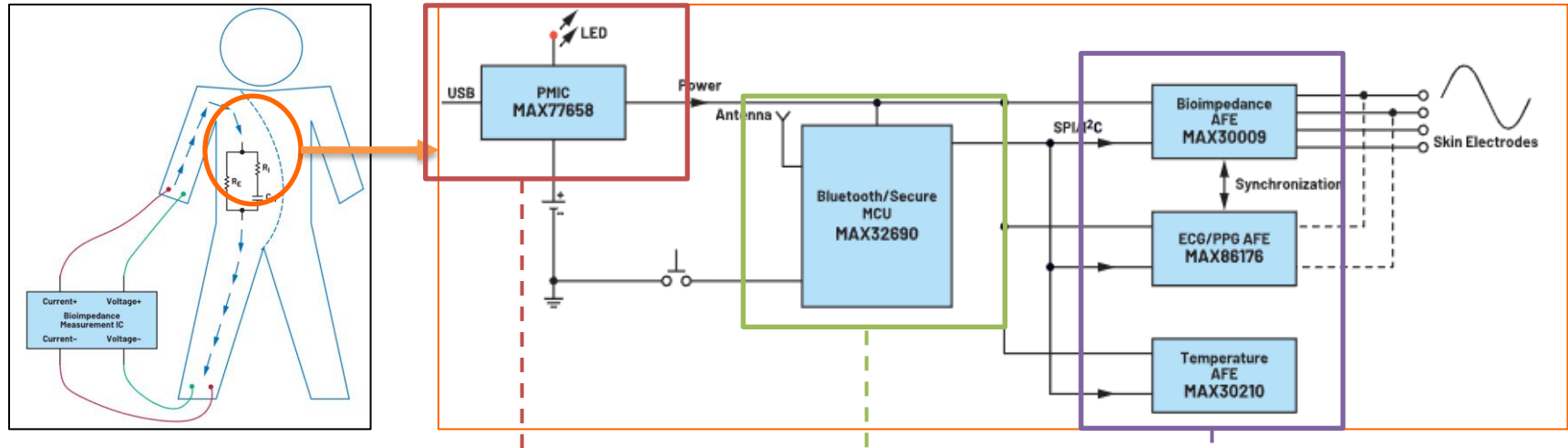
ICF and ECF are highly conductive,
can be equated to resistors

Cell membranes

Voltage difference exists across the
lipid bilayer like a capacitor



AFE for VSM device



PMIC

- Reduction of system size. Responsible for USB charging and providing stable voltage rails to the entire system

MCU

- controlling and configuring all AFE (enable and chip select) and coordinate AFEs (Propagation delay)

AFE

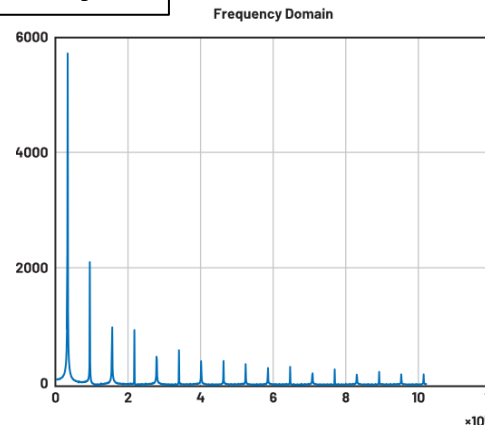
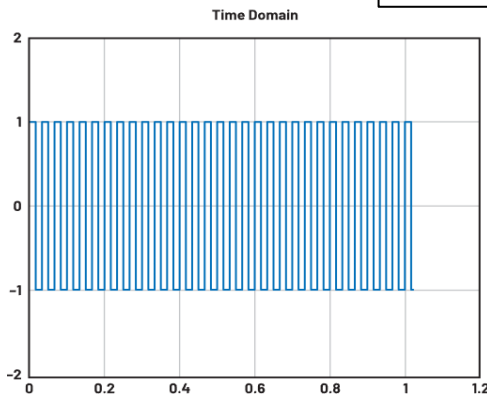
- The combined data from Bioimpedance, ECG, and PPG offers enhanced physiological reference value

Measurement

- $Z = V / I$ (Quadrature demodulation)
- By delivering an **electrical current** into the body and measuring the resulting voltage (V), the body's impedance can be calculated (V/I).

Current measurement solutions mostly use square wave pulse and Sine wave as excitation

Fourier Analysis



square wave is composition of many different frequency sine waves

Harmonics quickly decays ($1/n$) as frequency increases

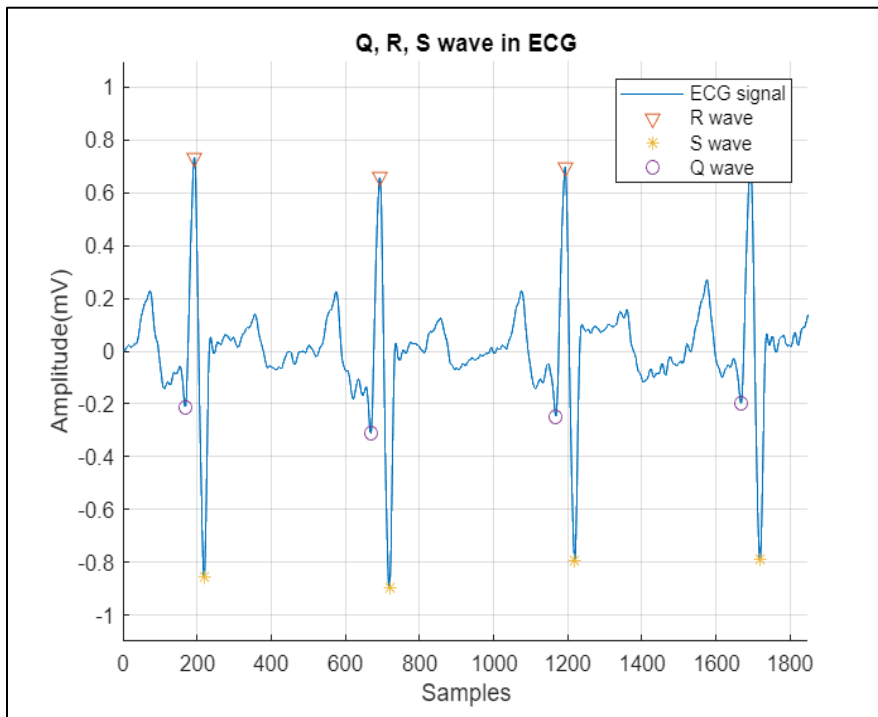
Our laboratory prototype employs a 100 kHz sine wave current as excitation signal.

ECG signal introduction

ECG can be divided into three main parts:



The results were obtained using IA, BRF, and BPF filtering.

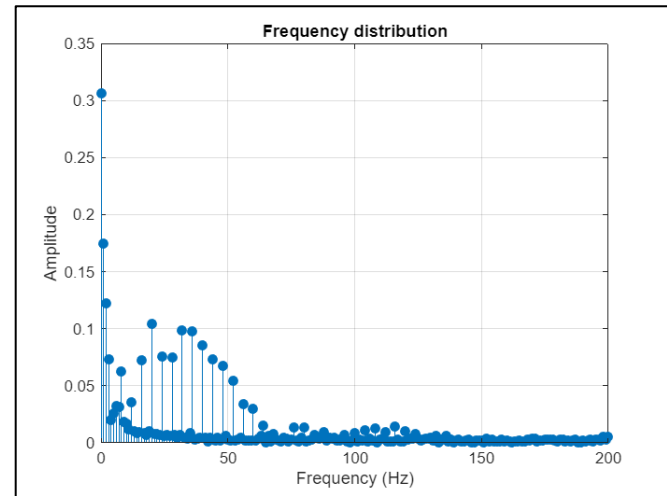
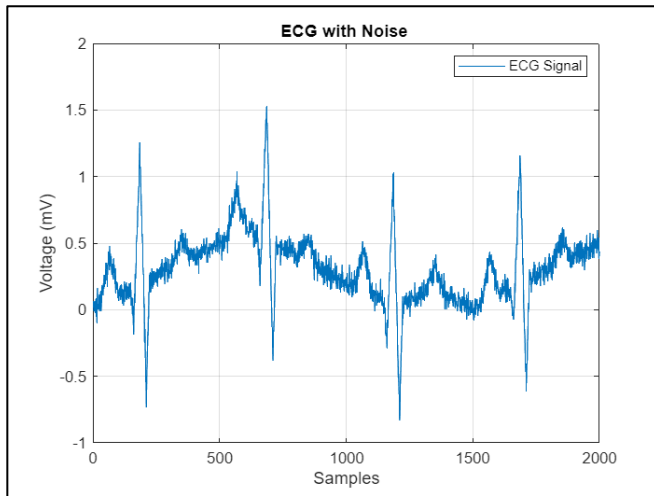


- P wave :
atrial depolarization
- QRS complex :
ventricular depolarization
- T wave :
ventricular repolarization

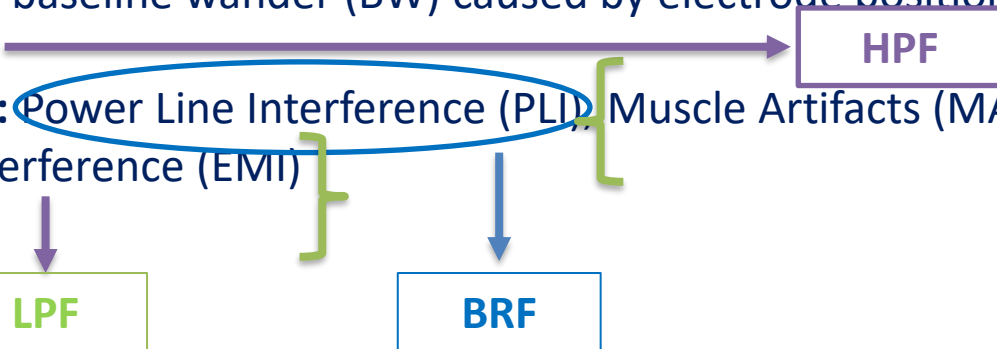
Simulation

- Origin Signal of ECG with noise

Fourier Analysis



- **Low-frequency noise** : baseline wander (BW) caused by electrode position and skin impedance
- **High-frequency noise** : Power Line Interference (PLI), Muscle Artifacts (MA), or Electromagnetic Interference (EMI)



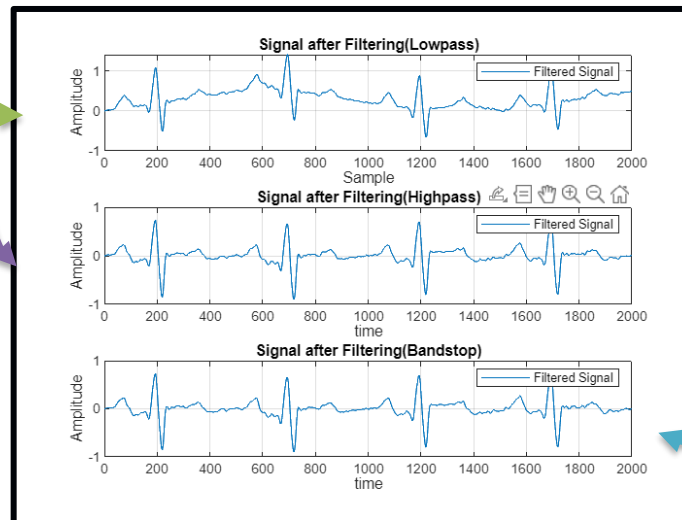
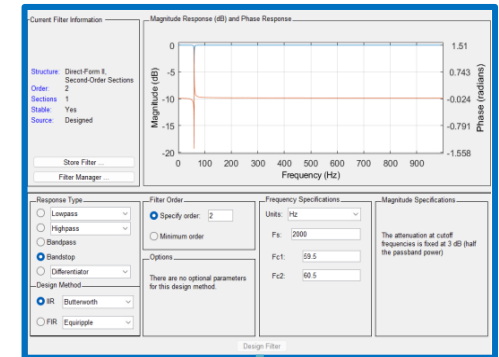
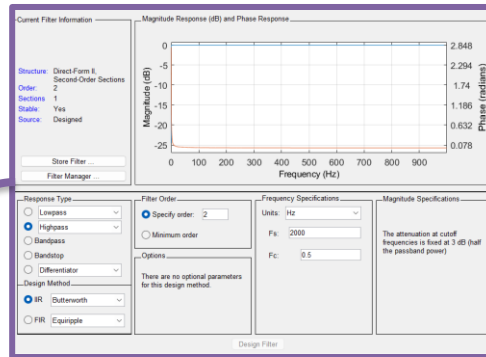
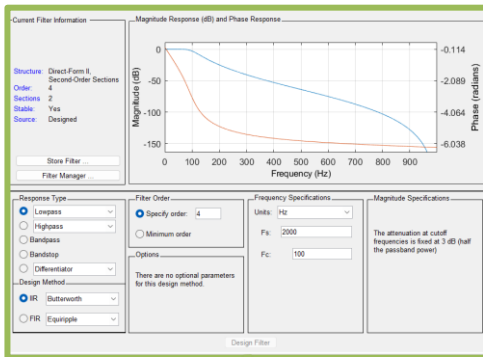
Simulation

- MATLAB Filter Designer

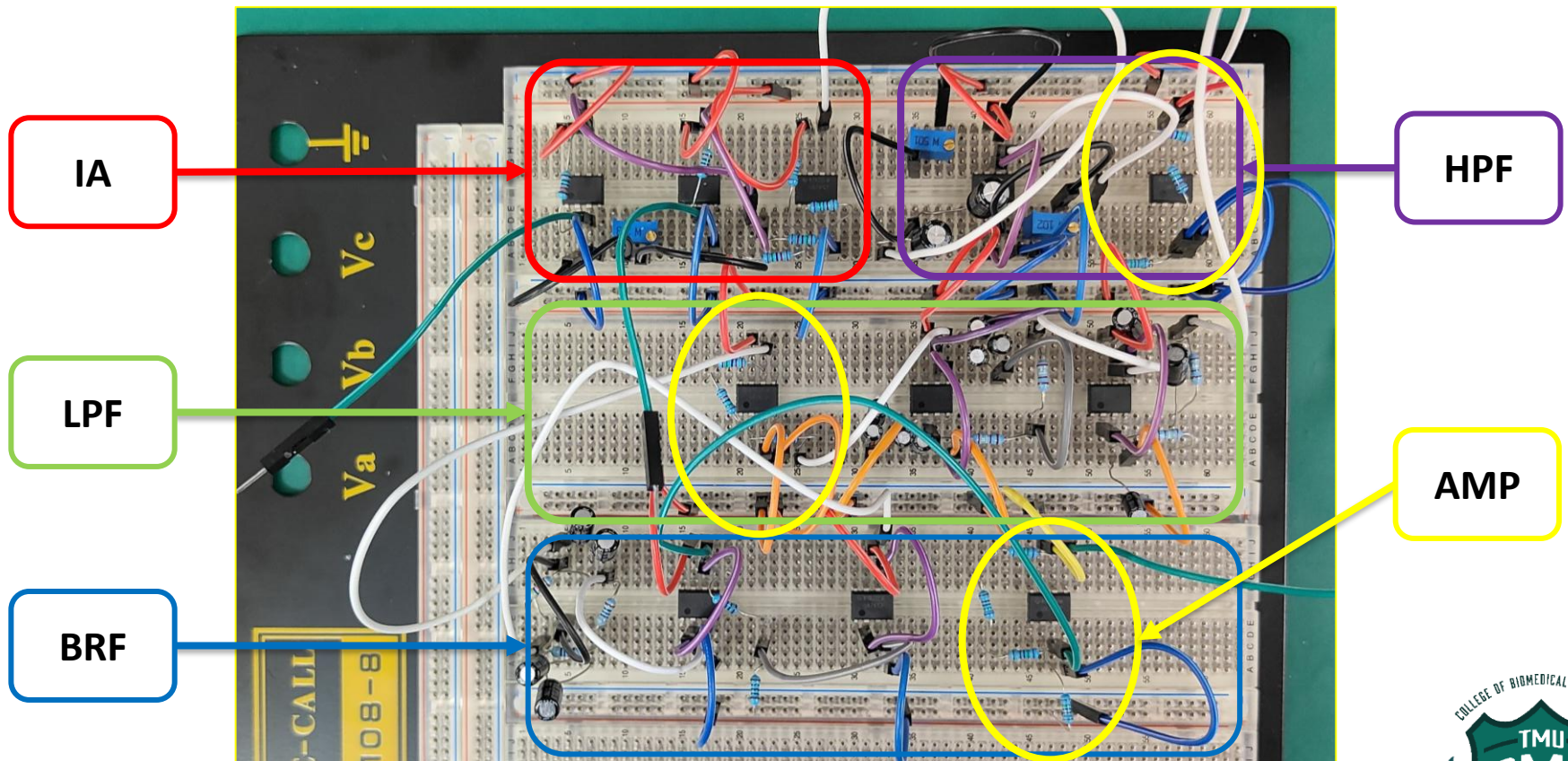
LPF

HPF

BRF



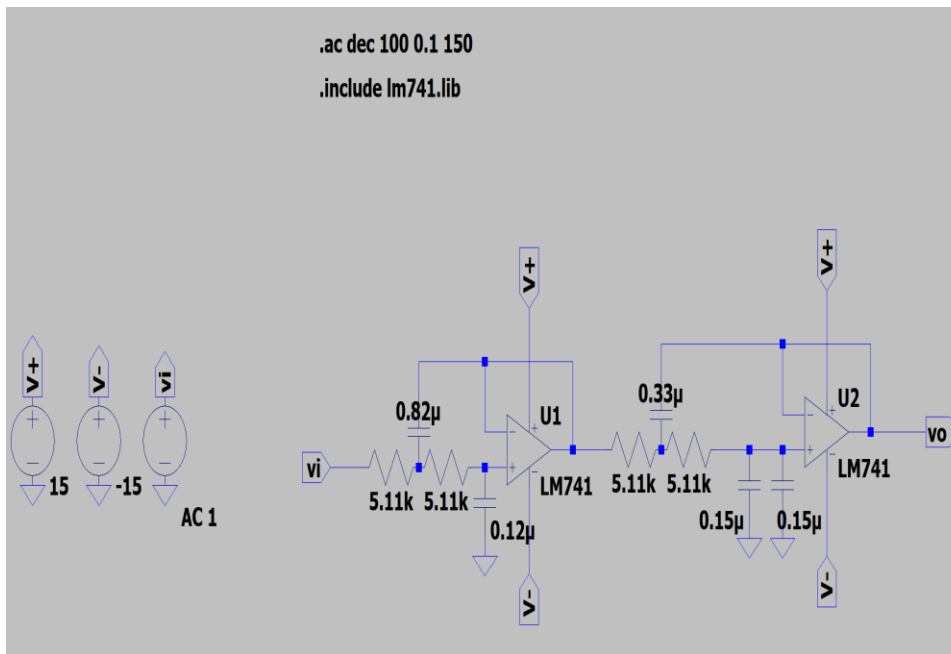
CKT Model Design



CKT Model Design

LPF

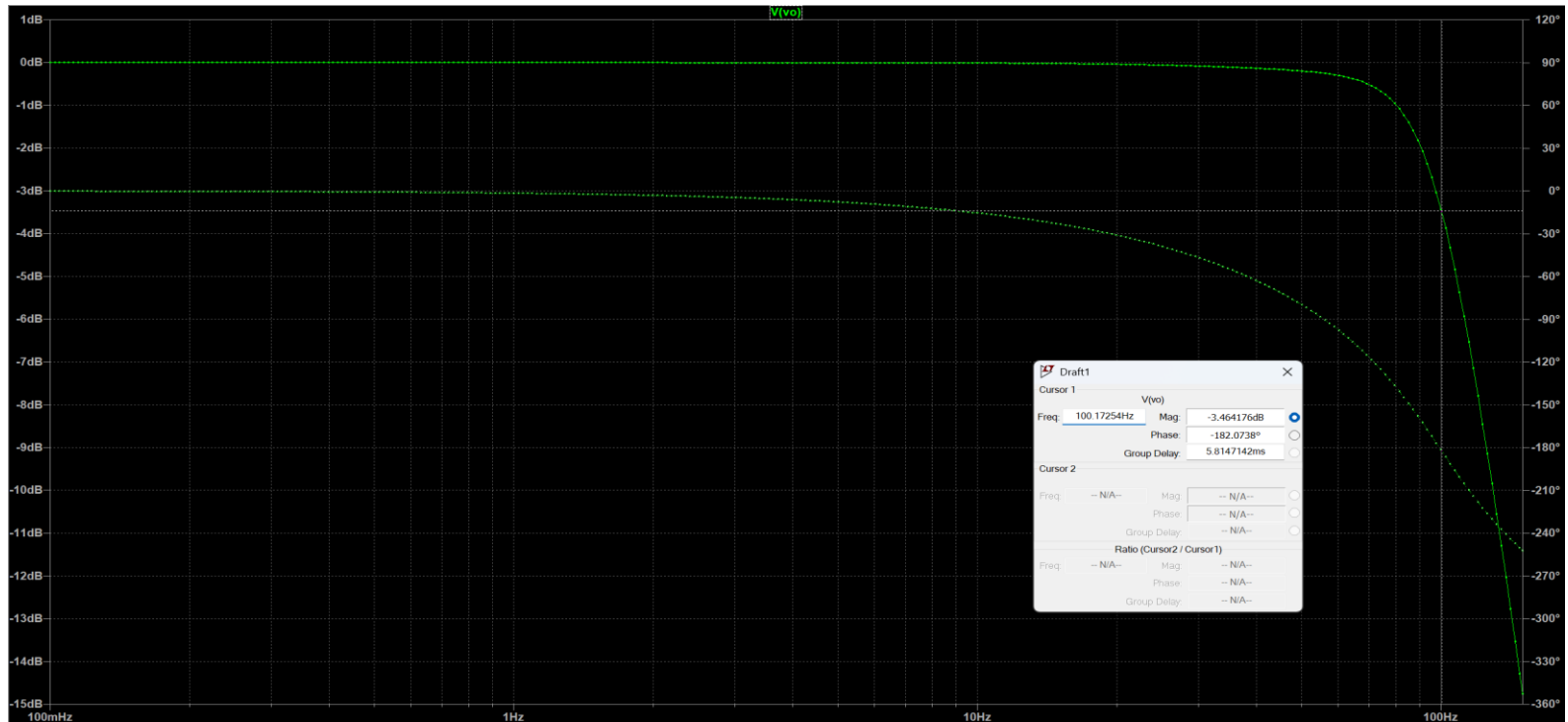
- Spice model : **Butterworth Lowpass Filter (4-Order)**
filter out high-frequency electromyogram (EMG) signals



- $f_c = 100 - 150$ Hz
- Scaling :
- set $R = 1$; $\omega_c = 1$ (Prototype)
- $$H(s) = \frac{1}{s^2 + \frac{2}{RC_1}s + \frac{1}{R^2C_1C_2}}$$

Result 1

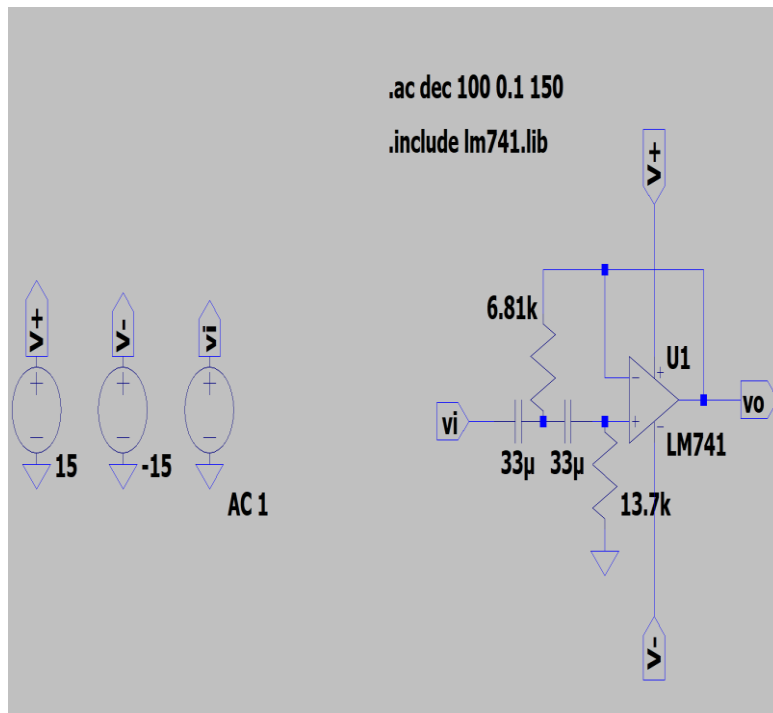
- Spice Simulation :



CKT Model Design

HPF

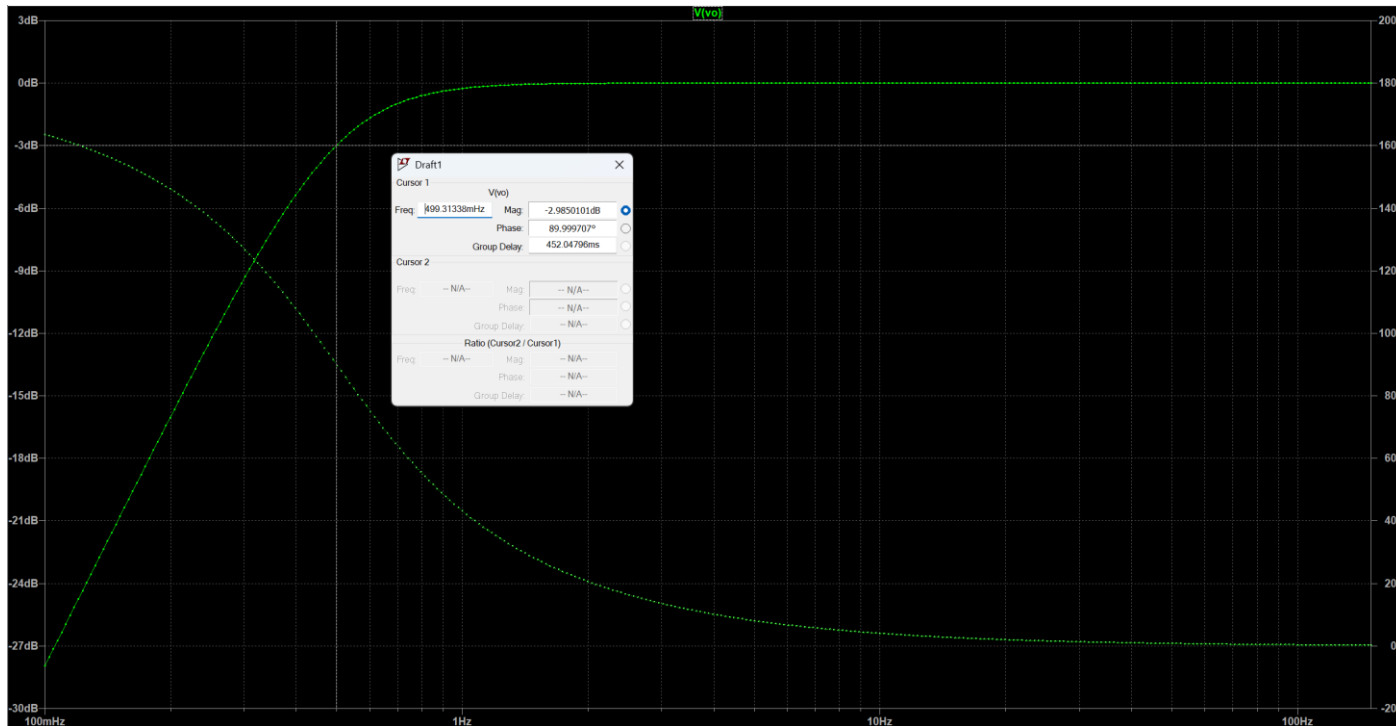
- Spice model : **Butterworth Highpass Filter (2-Order)**
Remove the DC offset



- $f_c = 0.05 - 0.5$ Hz
- Scaling : set $C = 1$; $\omega_c = 1$ (Prototype)
- $\omega'_c = k_f \omega_c$; $c' = \frac{1}{k_f k_m} c$; $R' = k_m R$
- $$H(s) = \frac{1}{s^2 + \frac{2}{CR_2}s + \frac{1}{C^2 R_1 R_2}}$$

Result 2

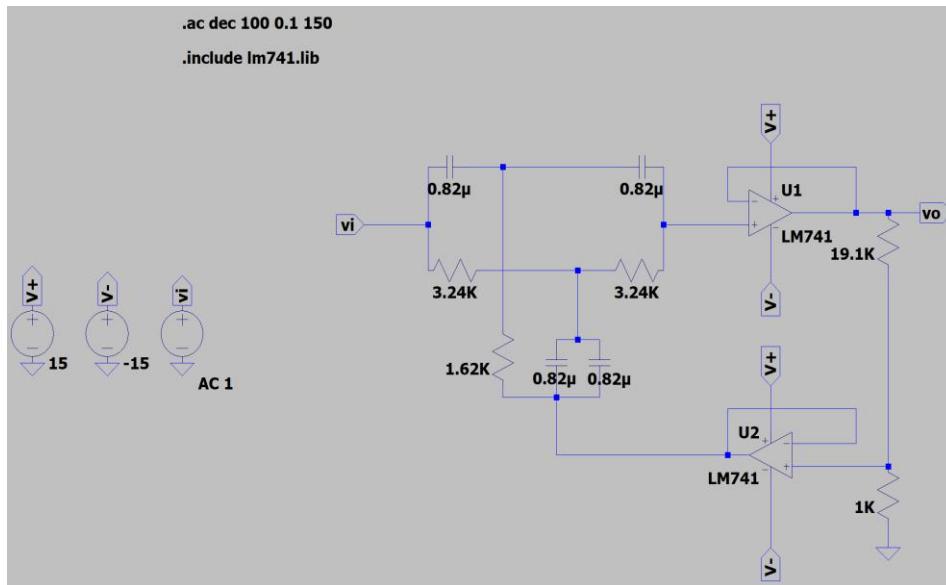
- Spice Simulation :



CKT Model Design

BRF

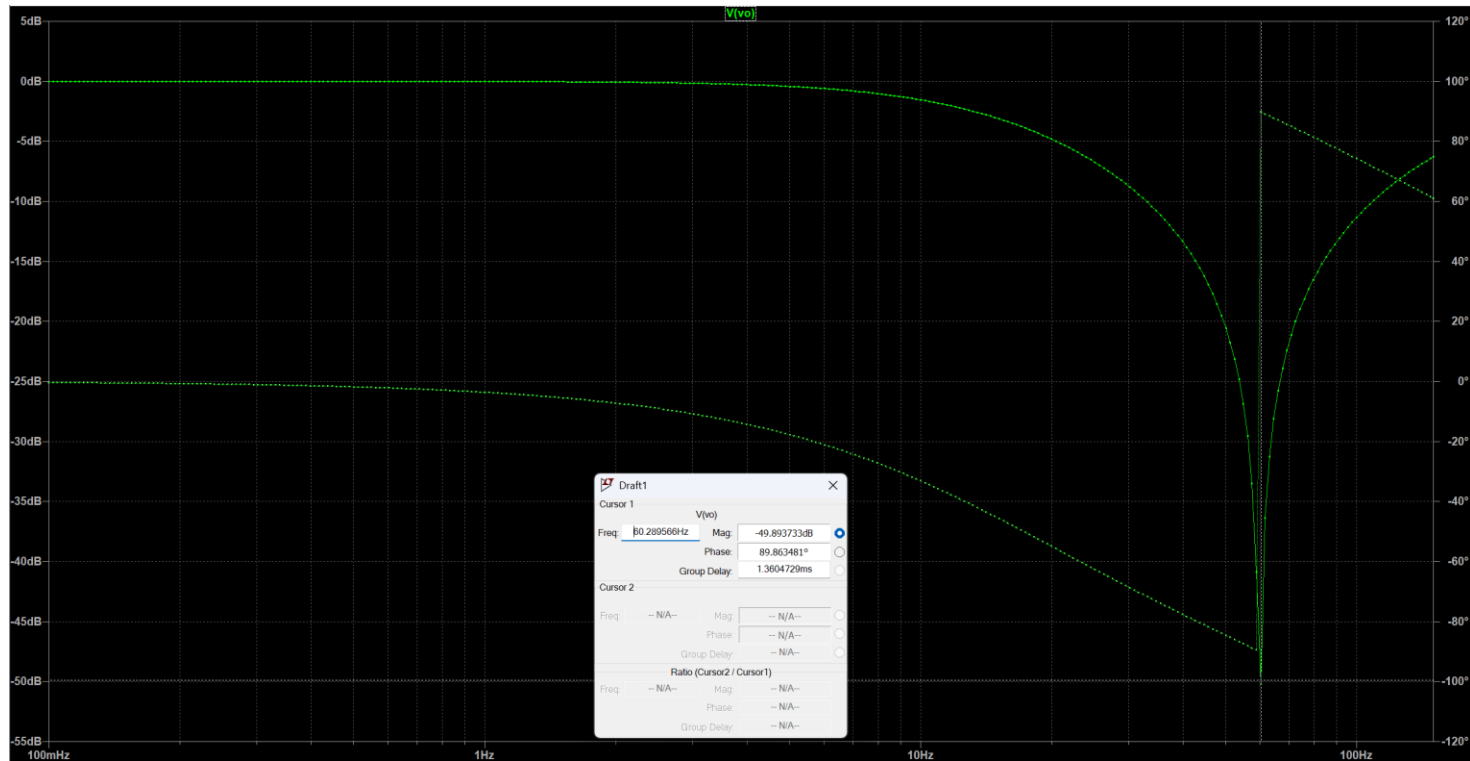
- Spice model : High Q Twin-T notch filter
Filtering the power input signal



- $f_0 = 50/60 \text{ Hz}$
- $H(s) = \frac{s^2 + \omega_0^2}{s^2 + \beta s + \omega_0^2}$
- $f_0 = \frac{1}{2\pi RC}$
- $\omega_0 = \frac{1}{RC}$
- $\beta = \frac{4(1-\sigma)}{RC}$
- $\sigma = 1 - \frac{1}{4Q}$

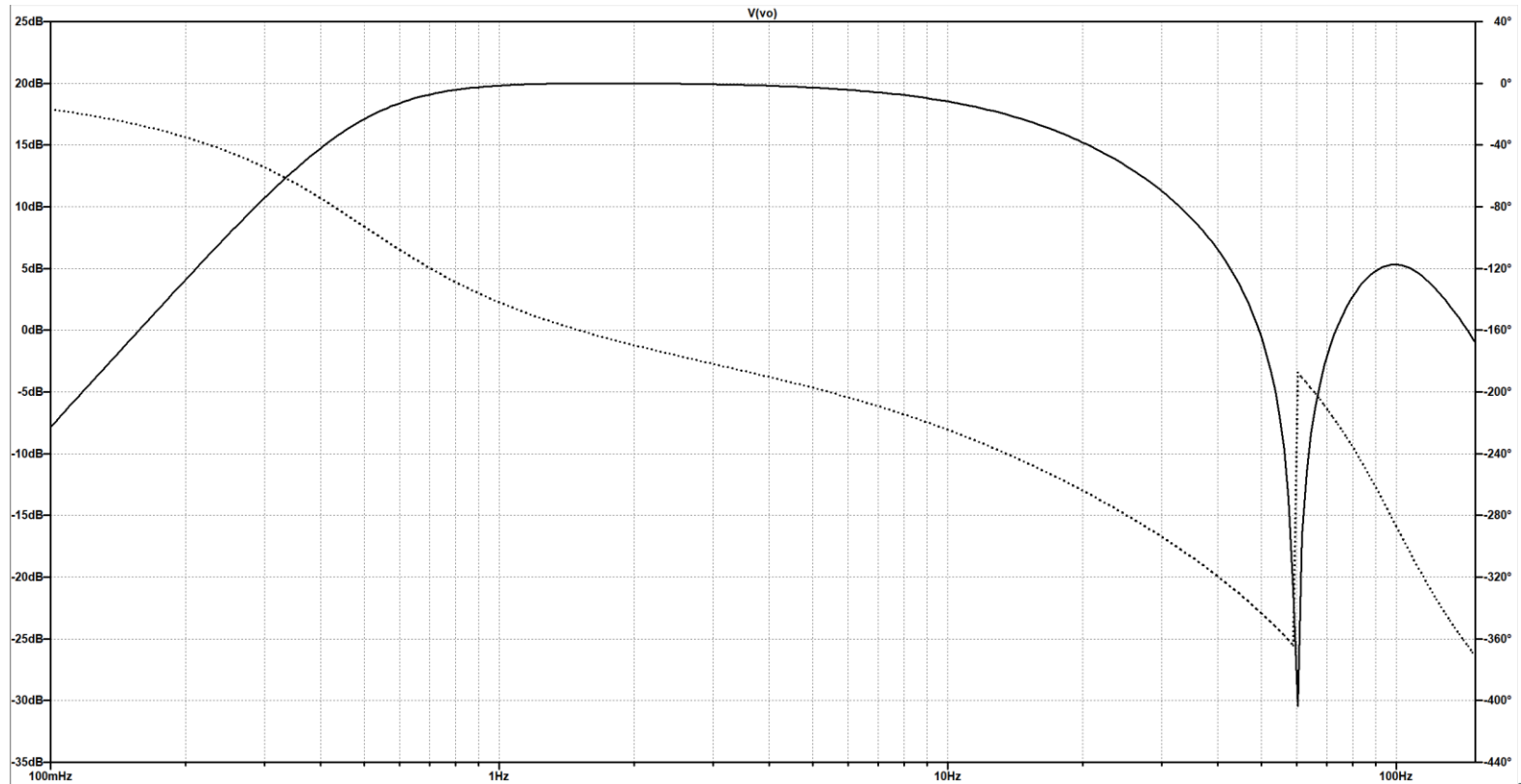
Result 3

- Spice Simulation :



System FRA

- Spice Simulation :



System FRA

- Actual:

